
CHAPTER 6

Conclusions

In this chapter we will discuss the final conclusions reached after the work has been completed. In contrast to the conclusions of the results, here we will focus on more general conclusions.

Getting back to the objectives set at the beginning of the work, we can say that we have managed to create a predictive model of football results based on Machine Learning. To say this, we rely on the comparison made between our predictive models and the benchmark used, the bookmakers' odds. We have obtained an overall accuracy equal to that obtained by bookmakers. Bookmakers have been doing this for years and have very well prepared working teams for this, as well as using more complete data than we had for this work. In addition, our model makes better predictions in the most difficult matches to predict, the draws. That is why we believe that the results obtained are satisfactory if we compare them with this Baseline.

In addition, we have tested two different classification Machine Learning algorithms and compared them with each other. We have seen how neural networks are algorithms that need precise configurations to obtain good results, as well as a proper feature engineering and an iterative process of model error analysis to find the way to calibrate the prediction distributions, especially ~~in a problem like this one~~ where the target classes are not evenly distributed. On the other hand, we have performed experiments with the Gradient boosting algorithm LightGBM, which has surprised us as it is possible to achieve really good results simply by optimising hyperparameters.

On the other hand, it is complicated to compare our work with the related works presented, since, as we have already mentioned, the fact of using totally different data or approaches (binary classification) complicates the comparison. However, we consider it appropriate to highlight, for example, the similar success obtained with the work carried out by Tax and Joulstra in which they predict the outcome (draw, home or away win) from thirteen seasons of the Dutch league also applying MLP and dimensionality reduction methods [Tax and Joulstra, 2015]. Other works related to outcome prediction are more difficult to compare with ours, as they use a completely different approach or data, both in the data used and in the type of matches to be predicted.

Our ANN-based model shows how data based on general statistics from previous matches can be used to make bookmaker-level predictions and improve the accuracy of the most difficult matches to predict, draws, which have so far been one of the biggest problems in result prediction. In our results we demonstrate something that previous works have not paid much attention to, the importance of not only using recent results, but also old

matches to give a complete match context to the model. In addition, we have proposed solutions, not presented so far in works on score prediction, to combat the problem of the unbalanced dataset that causes problems when predicting draws. With these solutions we have been able to predict, in many cases, close matches with greater accuracy and confidence.

Related to this, as a further objective, we have been able to draw from the exploratory and outcome analyses conclusions about certain hidden or unintuitive relationships that exist between certain statistics of previous matches and the outcome of a football match. These relationships have helped us to propose solutions to enhance the predictive performance of our models. We have already mentioned that counter-intuitively results or statistics, such as corners, from 1 or 5 years ago have a stronger relationship with the result of a current match than the result of the previous week's match. We have also seen that previous results between two teams do not usually have a decisive influence on the outcome of a match and therefore our models do not give them much importance.

In addition to the conclusions about the results obtained, we will discuss in the following section some findings about the relationship of our work with the courses taken during the degree.

6.1 Relationship between the work carried out and the studies undertaken

This work has required the application of many different fields that I have studied during my degree in Computer Engineering. In this section we will explain what knowledge I have acquired during the degree I have applied in this work as well as the new knowledge I have acquired during its realization.

On the one hand, I have been able to apply the knowledge acquired in software engineering during the implementation of the whole infrastructure of the experiments. Thanks to this I have been able to implement different types of neural networks by re-using code through class inheritance, which is a property of Object Oriented Programming. Moreover, thanks to the acquired knowledge related to Data Structures and Databases I have been able to implement the code of the experiments, both the execution of the experiment and the writing of logs and results, in a efficient way in terms of computational cost.

Furthermore, related to the Data Science knowledge, it has been very useful to have been taught during the degree Data Science concepts such as overtraining, the bias of a model, the optimisation of a Machine Learning model or statistical concepts such as the different continuous and discrete distributions or the statistical analysis of ANOVA. It has also been very useful to learn machine learning algorithms such as Multi-Layer Perceptron or K-Means..

In addition, during my academic exchange I had the opportunity to acquire a lot of valuable knowledge that I have been able to apply in this work. First of all, during that time I learned the theoretical and mathematical foundations behind neural networks and other Data Science related algorithms such as PCA. In addition, I learned to use Data Science tools (Pandas, Numpy, Matplotlib, Scikit-Learn, etc.) in a practical way that has helped me to speed up data processing and visualisation tasks. This work has helped me to reinforce and deepen the knowledge I acquired both theoretically and practically in Data Science and Software Engineering.

Apart from what I learnt during my degree, during this work I have developed new technical, methodological and work planning skills. On the technical side, I have learned to use some frameworks such as Optuna from scratch. Meanwhile, in the more methodological part, I have learned everything related to conducting a Data Science research work, from researching related work and the State-Of-The-Art to defining a methodology for experiments and analysis. I have also learnt to deal with situations where I was stuck and did not know how to proceed.

As in any learning process, problems and difficulties have arisen and I have had to face them and look for solutions. These aspects will be discussed in the next section.

6.2 Difficulties and limitations during my thesis

The difficulties and limitations that have arisen during the work have been basically two: difficulty in finding a suitable database to carry out the work and difficulty in automating the experiments.

Regarding the first point, we must be aware of the important limitation to obtain good results due to the fact of having such limited datasets, whether due to the amount of data (Event Based dataset) or due to the information contained in it (Historical dataset). This is why many hours have been invested not only in implementing the transformation of the original databases into trainable data, but also in thinking and designing the characteristics of the dataset based on the original data.

The other difficulty has to do with the development of the experiment infrastructure and its use. Moreover, the design and implementation of such infrastructure has not been trivial as it had a high level of complexity and it was not easy to carry out tests to check the correct functioning of the code. In addition, the design of the infrastructure had to be reconsidered several times, either because of important changes in the functionalities because the new implemented functionalities required it, e.g. new training logs. It has also been necessary to make major changes in the implementation due to the limitation of the virtual machine's memory, which on one occasion led to changing the way in which the logs were collected and the outputs of each model were saved.

Additionally, the machine on which the experiments were performed had problems during the experiments, being unavailable for certain periods of time, which affected the continuous development of the experiments. However, these problems were finally solved and the experiments could continue as normal.

Finally, as it is common when you are introduced to a new subject, difficulties arose related to theoretical concepts that were new to us. Especially in the application of techniques that we had not used so far, such as the statistical analysis of ANOVA for feature selection or the application of Batch Normalization to a neural network.

Since this is a work to be carried out within a limited time span, some solutions were proposed in the course of the project, but could not be implemented. In the following section we will discuss these aspects.

6.3 Future work

The lack of time meant that it was not possible to develop many of the ideas and new research approaches that had been considered during the course of this work.

One of these new routes of research was to change the type of problem set to predict outcomes from a classification problem to a regression problem. The idea was to approximate the exact outcome of the match with a predictive model (these could be the same as the ones used in the paper, ANN and LightGBM). In fact, some exploratory tests have been carried out but they have not been conclusive and it has been decided not to include them in the report. However, the results we were able to obtain from these tests suggested that this approach is quite promising.

We also considered the possibility of considering the data and the problem as a time series prediction problem, in which the statistics of each match would be data points of a time series of statistics of each team, and when predicting the outcome of a match, the time series of both teams would be taken into account. This idea has not been developed further due to the limited time we had and the complexity it has.

As next steps, we could develop a new software infrastructure that would allow the automation of predictions with current matches. Through web scrapping it could be possible to collect data from current matches and add them to our training dataset in order to predict upcoming matches.

In parallel to this, we would continue to improve the predictive model, adding more data (leagues, previous seasons, lower divisions, etc) and also rethinking the problem of the non-uniform distribution of results in the dataset in order to make better quality predictions.